

General Guidance

Motivation

- Spatial skills are mental skills associated with visualising and manipulating objects¹. Training spatial skills has been associated with improved performance in various STEM subjects².
- There is ongoing research into delivering maths lessons integrated with spatial reasoning in P4-7 classes across Scotland. For more information, see: <https://stemspaceproject.co.uk>. These lessons similarly aim to develop spatial skills in a computing science context.

Practical advice

- Lessons use either Scratch's online code editor (<https://scratch.mit.edu/>) or Raspberry Pi's Python code editor (<https://editor.raspberrypi.org/en-US/projects/blank-python-starter>). Both of these code editors are free and online – no downloads needed. These can be used on iOS for local authorities where pupils are allocated iPads.
- All lessons are intended to be run in a double block (around 90 minutes) and have approximate runtimes 70-90 minutes.
- Lessons are intended to be social. Encourage pupils to work in pairs/teams and discuss their ideas, particularly for more challenging activities.

Lesson theming

- Lessons follow a theme of attempting to model a real-life process in code. The table below illustrates these themes:

Lesson no.	Theme
1	"Giving instructions to computers"

¹ For more information about spatial skills see: Linn, M.C. and Petersen, A.C. (1985) 'Emergence and Characterization of Sex Differences in Spatial Ability: A Meta-Analysis', *Child Development*, 56(6), pp. 1479–1498. Available at: <https://doi.org/10.2307/1130467>.

² Engineering: Veurink, N.L. and Sorby, S.A. (2019) 'Longitudinal study of the impact of requiring training for students with initially weak spatial skills', *European Journal of Engineering Education*, 44(1/2), pp. 153–163. Available at: <https://doi.org/10.1080/03043797.2017.1390547>.

Computing science: Parkinson, J. and Cutts, Q. (2020) 'The Effect of a Spatial Skills Training Course in Introductory Computing', in *Proceedings of the 2020 ACM Conference on Innovation and Technology in Computer Science Education*. New York, NY, USA: Association for Computing Machinery (ITiCSE '20), pp. 439–445. Available at: <https://doi.org/10.1145/3341525.3387413>.

Mathematics: Lowrie, T. and Logan, T. (2023) 'Spatial Visualization Supports Students' Math: Mechanisms for Spatial Transfer', *Journal of Intelligence*, 11(6), p. 127. Available at: <https://doi.org/10.3390/jintelligence11060127>.

	This lesson is intended to encourage pupils to see similarities/differences in how they give instructions to a partner and to consider writing Scratch code as another way of giving instructions.
2	“Talking to computers” This lesson introduces print and input statements. Pupils will use these to create short “conversations”. Pupils should reflect on how close their programs feel to a real conversation.
3	“Naming things with computers” This lesson introduces variables. This may be quite a challenging lesson for pupils. Don’t be afraid to spend quite a bit of time with pupils making sure they have a usable mental model of how variables function. For pupils, being able to have a conceptual understanding of variables and their purpose is the most important takeaway from this lesson. Pupils should reflect on how variables allow us to use specific terms, rather than generic statements. They should consider if this makes their conversations feel more realistic.
4	“Making decisions with computers” This lesson introduces if/else statements and gets pupils to extend their conversation programs to make decisions about what to say in their conversations. Pupils should consider if the ability to “decide” what to say makes their conversation feel more realistic.
5	“Drawing with computers” Pupils will explore Python’s turtle module. This lesson is intentionally similar to lesson 1. Python turtle was chosen due to the similarity of its syntax to Scratch pen. Pupils should be encouraged to compare and contrast their experiences of using Scratch pen and turtle.
6	“Computational art critics” This lesson allows pupils some more freedom in terms of being able to design and create their own pieces of “Python

	artwork”. The planning aspect of this lesson is important and ensuring pupils follow the review process.
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Lesson structure

- The diagram below shows the typical structure of lessons.



- Lessons follow the idea of a semantic wave³. The key point to take from this is the importance of the final two steps where pupils link what they have learned during the lesson to what they see in real-world applications and evaluate what they have created.

Using the language transfer pedagogy

- Lessons use Tshukudu’s language transfer pedagogy⁴. Pupils are expected to have some prior knowledge of Scratch (basic blocks control the movement of Scratch, awareness of variables and if/else) before starting these lessons.
- In lesson 2, pupils will complete a so-called “guess quiz” which will require them to answer questions as a team about short sections of Python code. Pupils are not expected to know any Python prior to this activity – this activity is intended to see what knowledge from Scratch pupils are able to transfer to Python.
- Their quizzes do not have to be formally marked but it is useful to look at these to get an idea of what concepts pupils are able to transfer and what they might struggle with more.
- Tshukudu classifies concepts depending on their similarity or difference in a new language. Further information is provided in the document provided with the quiz which breaks down the intended classifications of the given questions. This document is for additional information only and is not necessary for delivering these lessons.

³ Curzon, P. *et al.* (2020) ‘Using semantic waves to analyse the effectiveness of unplugged computing activities’, in *Proceedings of the 15th Workshop on Primary and Secondary Computing Education. WiPSCE ’20: Workshop in Primary and Secondary Computing Education*, Virtual Event Germany: ACM, pp. 1–10. Available at: <https://doi.org/10.1145/3421590.3421606>.

⁴ Tshukudu, E., Cutts, Q. and Foster, M.E. (2021) ‘Evaluating a Pedagogy for Improving Conceptual Transfer and Understanding in a Second Programming Language Learning Context’, in *Proceedings of the 21st Koli Calling International Conference on Computing Education Research*. New York, NY, USA: Association for Computing Machinery (Koli Calling ’21), pp. 1–10. Available at: <https://doi.org/10.1145/3488042.3488050>.

Spatial skills

- The term “VPC” is used throughout. This stands for “Visualise-Predict-Check”. This is a process used to train spatial skills. In these lessons, this will usually involve showing pupils a section of code and asking them to visualise the execution. Pupils will then make a specific prediction about what the program will do/print out. Finally, the code will be run and pupils will reflect on whether or not their prediction was correct. For further information on VPC:
<https://link.springer.com/article/10.1007/s13394-020-00321-2>.
- Additionally, most coding activities require pupils to first write their code on paper, then run it and note any issues. This is intended to be a VPC process and writing their code on paper first is an important step as this gets pupils to visualise the code they are writing.
- Activities which encourage pupils to build a model of a process in code are also aimed at improving spatial skills. Encourage pupils to evaluate how close their model is to the real-life process.

AI usage

- It is recommended that there should be no AI involvement in these lessons. These lessons are intended to train mental skills in pupils and it is the author’s belief that AI usage would undermine this.